Points to Remember

- Heat is transferred from hot region to cold region.
- Heat is transferred in three forms: conduction, convection and radiation.
- Conduction takes place in solids and convection takes place in liquids and gases.
- Radiation takes place in the form of electromagnetic waves.
- There are three scales of temperature: Fahrenheit scale, Celsius or Centigrade scale and Kelvin or Absolute scale.
- Amount of heat energy absorbed or lost by a body is determined by three factors: mass of the body, change in temperature of the body, nature of the material of the body.
- The SI unit of specific heat capacity is $Jkg^{-1} K^{-1}$.
- Among all the substances, water has the highest specific heat capacity.
- SI unit of heat capacity is J/K.
- Depending upon the temperature, pressure and transfer of heat, matter is converted from one state to another.

A-Z GLOSSARY

Conduction	Process of transfer of heat in solids from a region of higher temperature to a region of lower temperature without the actual movement of molecules.
Convection	Flow of heat through a fluid from places of higher temperature to places of lower temperature by movement of the fluid itself.
Radiation	Flow of heat from one place to another by means of electromagnetic waves.
Temperature	It is the degree of hotness or coolness of a body.
Specific heat capacity	The amount of heat required to raise the temperature of 1 kg of the substance by 1°C or 1 K.
Heat capacity	The amount of heat energy required to raise the temperature of a body by 1°C.
Melting or fusion	Process in which a solid is converted to liquid by absorbing heat.
Freezing	Process in which a liquid is converted to solid by releasing heat.
Vaporization	Process in which a liquid is converted to vapour by absorbing heat.
Condensation	Process in which a vapor is converted to liquid by releasing heat.
Latent heat	Amount of heat energy absorbed or released by a substance during a change in its physical state without any change in its temperature.
Specific latent heat	Amount of heat energy absorbed or liberated by unit mass of substance during change of state without causing any change in temperature.

TEXT BOOK EXERCISES

7PZ56

I. Choose the correct answer.

1. Calorie is the unit of

   a) heat b) work

   c) temperature d) food

2. SI unit of temperature is

   a) fahrenheit b) joule

   c) celsius d) kelvin

IX_SCI_EM_Unit-07_PHY.indd 88 12/7/2022 6:13:32 PM